

CO5 AT2 SHORT ANSWER

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QUESTIONS

Q1 ✓
How many edges must a Minimum Spanning Tree of a graph with V vertices have?
2 marks

Q2 ✓
In a DAG, which node will always have an in-degree of zero in a topological ordering?
2 marks

Q3 ✓
Differentiate between directed and undirected graphs.
2 marks

Q4 ✓
Define a connected graph.
2 marks

Q5 ✓
What is a topological sort?
2 marks

5/5 answered

Q1 How many edges must a Minimum Spanning Tree of a graph with V vertices have? 2 marks

How many edges must a Minimum Spanning Tree of a graph with V vertices have?

Your Answer

A Minimum Spanning Tree of a graph with V vertices must have exactly $V - 1$ edges.

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2 marks

Q5 ✓
What is a topological sort?
2 marks

5/5 answered

Q2 In a DAG, which node will always have an in-degree of zero in a topological ordering? 2 marks

In a DAG, which node will always have an in-degree of zero in a topological ordering?

Your Answer

In a DAG, the node that always has an in-degree of zero in a topological ordering is the source node (also called a starting node). Since it has no incoming edges, it cannot depend on any other vertex, so it must appear first in every valid topological sort.

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QUESTIONS

Q1 ✓
How many edges must a Minimum Spanning Tree of a graph with 5V vertices have?
2 marks

Q2 ✓
In a DAG, which node will always have an in-degree of zero in a topological ordering?
2 marks

Q3 ✓
Differentiate between directed and undirected graphs.
2 marks

Q4 ✓
Define a connected graph.
2 marks

Q5 ✓
What is a topological sort?
2 marks

5/5 answered

Q3 Differentiate between directed and undirected graphs.

2 marks

Differentiate between directed and undirected graphs.

Your Answer

In an undirected graph, edges have no direction, meaning they represent a bidirectional relationship between two vertices (e.g., an edge between A and B is the same as between B and A). In a directed graph, edges have a specific direction, represented by arrows, indicating a one-way relationship from a source vertex to a target vertex.

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QUESTIONS

Q1 ✓
How many edges must a Minimum Spanning Tree of a graph with 5V vertices have?
2 marks

Q2 ✓
In a DAG, which node will always have an in-degree of zero in a topological ordering?
2 marks

Q3 ✓
Differentiate between directed and undirected graphs.
2 marks

Q4 ✓
Define a connected graph.
2 marks

Q5 ✓
What is a topological sort?
2 marks

5/5 answered

Q4 Define a connected graph.

2 marks

Define a connected graph.

Your Answer

A connected graph is an undirected graph in which there is a path between every pair of vertices. In other words, there are no isolated vertices or separate components within the graph.

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QUESTIONS

Q1

How many edges must a Minimum Spanning Tree of a graph with V vertices have?
2 marks

**Q2**

In a DAG, which node will always have an in-degree of zero in a topological ordering?
2 marks

**Q3**

Differentiate between directed and undirected graphs.
2 marks

**Q4**

Define a connected graph.
2 marks

**Q5**

What is a topological sort?
2 marks



5/5 answered

Q5 What is a topological sort?**2 marks**

What is a topological sort?

Your Answer

A topological sort of a directed acyclic graph (DAG) is a linear ordering of its vertices such that for every directed edge $u \rightarrow v$, vertex u comes before v in the ordering.



 Save Answer